

Finding Your Project

*Who are you talking to,
and what are you saying to them?*



What is this in conversation with?

- Persian folklore and the 1001 Nights tradition
- Wittgenstein's philosophy of language
- Interactive fiction and narrative games
- Generative AI research (CHI, AIIDE papers)
- Feminist game design practice

The question your proposal has to answer:

What is **your project** in conversation with?

Not just what you're making — but whose prior work you're responding to, what debate you're entering, what tradition you're working within or against. This is what distinguishes Masters-level research from a personal project.

Before we go further:

Two distinctions that matter

Artefact type ≠ Discourse register

ARTEFACT TYPE

What you make

- Software / game
- Written dissertation
- Portfolio of creative work
- Live or participatory work
- AI / generative system
- Other

E.g. : live / participatory work is an artefact type, not a register — the same LARP can be art-world facing, part of a media campaign, or a design research output.

≠

DISCOURSE REGISTER

Who you're talking to, and how

From sociolinguistics: register describes how language and presentational choices signal community membership and situate the speaker within a discourse. Not just a genre or a discipline — a set of conventions that mark you as belonging to a particular community of practice.

- Academic game studies
- Art world / gallery practice
- Auteur / independent games
- Technical / systems research
- Design research / HCI
- ...and others

Why register matters beyond the project

Your choice of discourse register is also a statement about professional identity and career strategy.

Funding

Arts Council, AHRC, UKRI, EU Horizon, industry R&D grants, commissions — each belongs to a different register. You apply to the funders whose criteria match your register.

Quality criteria

What counts as excellent work differs by register. A technically flawless but aesthetically inert game is excellent in one register and a failure in another.

Career trajectory

Academic positions, gallery representation, studio employment, independent practice — each requires fluency in a different register. Your CV tells a register story.

Peer community

Who reviews your work, who you submit to, whose opinion you seek — all determined by register. Discourse communities are professional networks.

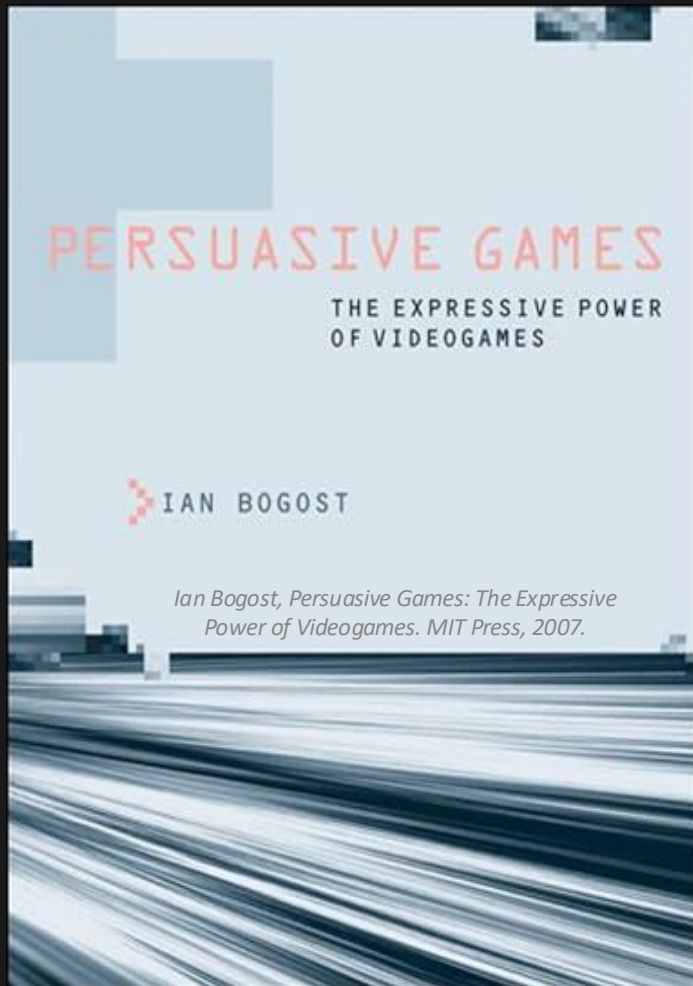
The proposal

Knowing your register lets you write a sharper contextual review, a more specific research question, and a more honest account of what success looks like.

A survey of:

Discourse Registers in Games and Creative Technology

These are not exhaustive, and not mutually exclusive.



Register 1

Academic Game Studies

Bogost, Persuasive Games (2007) — or Jayemanne, Fassone

Addresses

Scholars in game studies, media studies, humanities — assumes familiarity with critical theory and academic conventions

Circulates via

MIT Press, university press monographs, Game Studies journal, Games and Culture, DiGRA, FDG proceedings

Quality criteria

Argumentative rigour, originality of claim, engagement with existing literature, peer review

Career here

Academic position, research grants, conference circuit, journal editorships

Conventions

Abstract, literature review, citation apparatus, peer review, academic press or journal publication



Eddo Stern, *Waco Resurrection* (2003).

Register 2

Art World / Gallery Practice

Eddo Stern — Waco Resurrection (2003); also Blast Theory

Addresses

Gallery audiences, peer artists, art critics and curators — assumes cultural literacy, not game literacy

Circulates via

Galleries, Ars Electronica, transmediale, Sundance New Frontier, art press, museum collections, artist residencies

Quality criteria

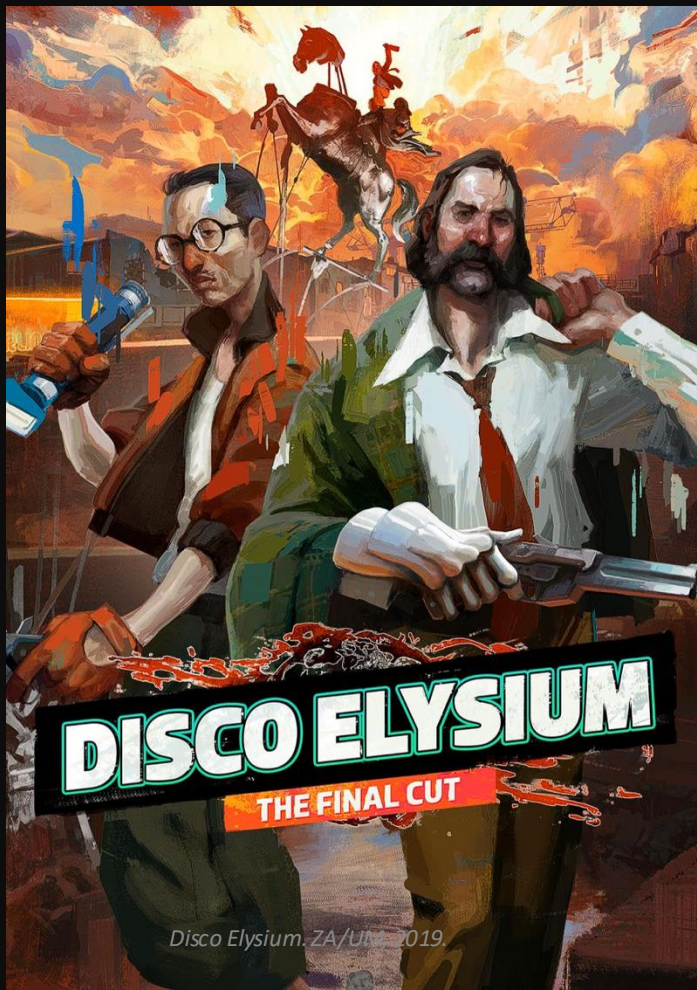
Conceptual originality, critical reception, exhibition history, artist statement, contribution to art discourse

Career here

Commissions, Arts Council and EU research funding, gallery representation, residencies — Blast Theory is exemplary

Conventions

Artist statement, exhibition documentation, critical essay, press release, documentation as practice



Register 3

Auteur / Independent Games

Disco Elysium — ZA/UM, 2019

Addresses

Players — specifically a literary and aesthetically sophisticated segment of the indie market. Not addressed to academics, even if available to scholarly analysis.

Circulates via

Steam, game press (Waypoint, RPS, Eurogamer), GOTY lists, cultural journalism, word of mouth

Quality criteria

Critical reception, player engagement, cultural impact, design coherence, artistic ambition

Career here

Indie studio, self-publishing, crowdfunding, publisher partnerships, GDC postmortems

Conventions

Trailer, press kit, Steam page, reviews, design postmortem — not academic citation

Key distinction

Available to game studies as an object of analysis — but not addressed to it. A dissertation about *Disco Elysium* talks to scholars. *Disco Elysium* talks to players.

WaveFunctionCollapse is Constraint Solving in the Wild

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ABSTRACT

Maxim Gumin's WaveFunctionCollapse (WFC) algorithm is an example-driven image generation algorithm emerging from the craft practice of procedural content generation. In WFC, new images are generated in the style of given examples by ensuring every local window of the output occurs somewhere in the input. Operationally, WFC implements a non-backtracking, greedy search method. This paper examines WFC as an instance of constraint solving methods. We trace WFC's explosive influence on the technical artist community, explain its operation in terms of ideas from the constraint solving literature, and probe its strengths by means of a surrogate implementation using answer set programming.

CCS CONCEPTS

• Theory of computation → Constraint and logic programming: Random walks and Markov chains; • Applied computing → Media arts: Fine arts; • Mathematics of computing → Solvers;

KEYWORDS

constraint solving, procedural content generation, texture synthesis

ACM Reference format:

Isaac Karth and Adam M. Smith, 2017. WaveFunctionCollapse is Constraint Solving in the Wild. In *Proceedings of FOG'17*, Hyannis, MA, USA, August 14-17, 2017, 10 pages.
<https://doi.org/10.1145/3102071.3110566>

1 INTRODUCTION

Constraint solving is neither a traditional nor well-known approach to procedural content generation (PCG). Nevertheless, this approach can be surprisingly effective for building controllable content generators. (Examples are discussed in section 3.) This paper examines the WaveFunctionCollapse (WFC) algorithm, an example-driven image generation algorithm recently developed by game developer Maxim Gumin, and illuminates it through the lens of constraint solving.

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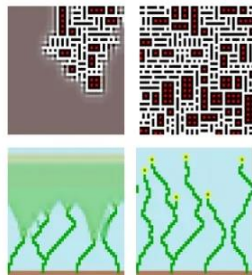


Figure 1: The WaveFunctionCollapse generator in action. The unresolved information in the images on the left is shown as the average color value of their possible outputs.

In 2016 and 2017, WaveFunctionCollapse attracted the attention of several indie and hobby game makers via social media.¹ Animations of the algorithm demonstrated not just the primary output, a large image generated in the style of a smaller artist-provided source, but also a surprisingly human-like mode of incremental creation. Referring to one of the major components of the algorithm, Gumin writes [12]: "I noticed that when humans draw something they often follow the minimal entropy heuristic themselves. That's why the algorithm is so enjoyable to watch." The animated visualizations, with the results gradually resolving themselves out of a fog of possibilities (Fig. 1), instantly showed that the algorithm works differently than familiar constructive or artifact-at-a-time generate-and-test methods.

The approach used draws deeply from techniques known in computer graphics for texture synthesis (we examine this literature more in a later section). Where WaveFunctionCollapse departs from

¹For example, Maxim Gumin's first WFC tweet "Procedural generation from a single example by wave function collapse" <https://github.com/imagima/WaveFunctionCollapse> ... (Which has hundreds of retweets)

<https://twitter.com/ElCrimino/status/781833475864277680>

and Danny Wyne's "3d tile placement with WFC" This algorithm is amazing. Inspired by @ElCrimino and based on @ElCrimino work <https://twitter.com/ElCrimino/status/781833475864277680>

<https://twitter.com/dwyne/status/810106761270243328> and others mentioned below.

Maxim Gumin, *WaveFunctionCollapse* (2016).
github.com/mxgmn/WaveFunctionCollapse
Karth & Smith, "WFC is Constraint Solving in the Wild",
AIIDE 2017.

Register 4

Technical / Systems Research

Wave Function Collapse — Maxim Gumin (2016)

Formalised: Karth & Smith, AIIDE 2017

Addresses

Researchers and engineers — assumes technical literacy, familiarity with algorithms and evaluation methodology

Circulates via

AIIDE, IEEE Transactions on Games, SIGGRAPH, CHI, arXiv preprints, GitHub repositories

Quality criteria

Reproducibility, measurable improvement over baselines, clarity of documentation, reusability, validity of evaluation

Career here

Research scientist, industry R&D, academic CS or AI position — often moves between academia and studio

Conventions

Technical paper with benchmark comparisons, open-source repository, README, version control history, supplementary materials

The register space is larger than this

Other registers worth knowing about:

Design research / HCI

CHI design track, Design Studies, Transactions on CHI — artefact as design argument; quality measured by contribution to design knowledge

Education / serious games

UKRI, Wellcome, educational publishers — quality measured by learning outcomes, pedagogical validity, institutional impact

Criticism / journalism

Waypoint, RPS longform, Hazel Monforton, Eurogamer features — addressed to a player-adjacent public, quality measured by insight and writing

Industry R&D

GDC talks, technical blogs, studio white papers — commercial accountability rather than scholarly peer review; high practical impact

Your register choice is partly a discovery — what community does this work naturally belong to? — and partly a decision — which community do I want to be part of? Both matter.

The most interesting work often sits at the boundary

Register crossing is not a failure of clarity — it can be a deliberate and generative intellectual position.

1001 Nights — Ada Eden

Art world / experimental practice + Technical AI research (CHI, AIIDE papers)

Funded through indie channels, exhibited in galleries, published academically. Each register demands different fluency.

Jeff Howard — Quests (2008)

Academic game studies (MIT Press monograph) + Practitioner discourse (designers actually read and use it)

A scholarly argument that also functions as a design resource. The crossover is intentional and constitutes the book's value.

Blast Theory — Can You See Me Now? (2001–)

Art world / gallery practice + Design research + EU research funding frameworks

The same project is submitted to gallery programmes and to academic conferences. Blast Theory maintain fluency in both.

But: you need to know you're doing it, and you need to speak fluently in both registers. The proposal is where you demonstrate that awareness.

From register to

Your Discourse Community

What is a discourse community?

Not a discipline. Not a genre. Not a programme.

- **The people whose opinion of your work you care about**
- Whose prior work you are responding to
- Whose conventions you are working within — or deliberately against

Choosing which community you're addressing is a creative and intellectual decision — it shapes what counts as a good research question, what counts as evidence, and what counts as success. It is also, as we've seen, a professional decision with long-term consequences.

Why this matters for your proposal

Your contextual review

Isn't just a reading list — it's a map of who came before you and why your work responds to them. It only makes sense relative to a discourse community.

Your research question

Should be answerable by the methods that community recognises, and interesting to them if answered. 'I want to explore X' is not a research question.

Your quality criteria

Differ by register. Different discourse communities have different priorities and different vocabularies for describing them.

Your process documentation

Should make the intellectual development of your work legible to an assessor — and legibility means shared conventions.

Now:

Positioning Your Project

Take 10 minutes

Three questions — write your answers

1 What are you making?

One sentence. A description, rather than a pitch. What is the thing, and what does it do?

2 Whose work does yours respond to?

Two or three names, works, or bodies of work you admire, react against, or build upon. They don't have to be games.

3 What would success look like in their company?

If your work were sitting alongside the work you named in Q2 — what would make it worthy of that conversation? What would have to be true of it?

Share back

A few volunteers — let's look at Q3 together.

- The question is not whether the project is good
- The question is whether you can locate it in a conversation
- If you can't yet — that's fine. That's what the contextual review and the workshops are for.

Your supervisor's job, and the workshops' job, is to help you get to a clear answer to Q3.

Back to the opening question:

What is 1001 Nights in conversation with?

You can now answer that. The next step: ask it of your own project.

Next session (Tuesday 2 June — LEC 2): given what you're saying and who you're saying it to — how do you make the work, and how do you know when it's working? We'll also begin to think about the relationship between discourse communities and the methodologies they deploy — which is not always a straightforward one.

Workshops begin tomorrow — Wednesday 27 May. Bring your proposal form.

FAITH + DOUTH
WORKSHOPS

